

Charge and Mass Considerations for Plasma Velocity Measurements in Rotating Plasmas

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Abstract Velocity of hydrogen plasmas rotating due to imposed $\mathbf{E} \times \mathbf{B}$ fields at the Maryland Centrifugal Experiment (MCX) (Ellis et al., Phys Plasmas 12:055704, 2005), where $\mathbf{E}$ is the electric field in the radial direction and $\mathbf{B}$ the magnetic field in the axial direction of a cylindrical configuration, has traditionally been measured using Doppler shifts of atomic spectra from impurity elements such as carbon. Ideally, the gyrocenter of trace particles rotates at the bulk plasma velocity, regardless of the charged state or trace particle mass. However, for sufficiently large applied $|E/B|$ (or equivalently, a sufficiently large ratio of bulk plasma rotation frequency and particle gyrofrequency), charged particles may have gyroradii that depart significantly from quasi-circular orbits drifting about the $\mathbf{B}$ field axis. This effect is investigated numerically with a single particle code that includes scattering, as well as experimentally at MCX. Numerical findings are compared to experimentally measured Doppler shifts of singly ionized helium and oxygen, and doubly ionized carbon atoms.

Keywords Centrifugal confinement · Particle orbits · Doppler spectroscopy

Introduction

Supersonic rotating plasmas are studied at the Maryland Centrifugal Experiment (MCX) [1] with a variety of

diagnostics and techniques. These studies are aimed at understanding velocity-shear stabilization of interchange magnetohydrodynamic (MHD) modes in a magnetic mirror configuration in which rotation is induced by the application of a radial electric field inside the mirror, thus creating $\mathbf{E} \times \mathbf{B}$ drift in the azimuthal direction. The magnitude of this azimuthal rotation is $v_\phi = |E/B|$. A schematic diagram of the MCX configuration is shown in Fig. 1.

In the context of MHD theory, v_ϕ represents the gyrocenter velocity of charged particles moving perpendicular to the magnetic field. For these particles, angular frequency about the gyrocenter is $\Omega_i = qB/m$ (where q is the particle's charge, m its mass, and B the magnitude of the magnetic field) and gyroradius $r_L = v_\perp/\Omega_i$ (i.e., the Larmor radius, where $v_\perp$ is in general not equal to v_ϕ). If $r_L \ll R$, where R is the radius at which the gyrocenter drifts with a velocity v_ϕ , then exact orbital dynamics may be neglected and the particle's trajectory approximated by that of its gyrocenter. Equivalently, q/m is neglected in this case.

Suppression of interchange modes is predicted to improve with increasing v_ϕ for centrifugally confined plasmas [2, 3]. It is thus of interest to investigate high $|E/B|$ ratios. Traditionally, *C III* (i.e., doubly ionized carbon) impurities have been used to estimate bulk plasma velocity [4, 5]. In this article we present preliminary studies of a single particle code that is used to trace orbits for which the $r_L \ll R$ assumption breaks down (“Particle Orbit Model”). Of particular interest are *He II*, *O II* (i.e., singly ionized ions), and *C III* ions since q/m is different for all and determines different orbit details under high $|E/B|$ ratios. Simulations are compared to plasma experiments at MCX (“Comparison Between Experimental Measurements and Simulations”). Discussion of results and future plans are presented in “Discussion”.

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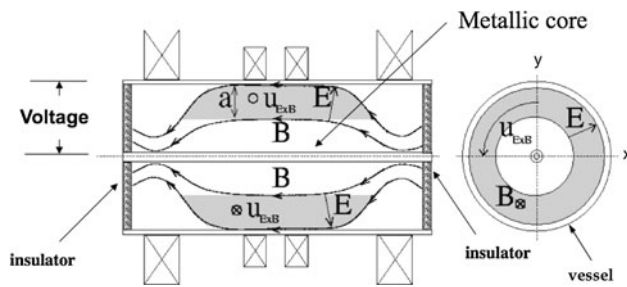


Fig. 1 Schematic of the MCX concept. The magnetic field is azimuthally and axially symmetric

Particle Orbit Model

The equation that describes the classical movement of a charged particle under electric and magnetic fields is $m\mathbf{a} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$, where $\mathbf{a}$, $\mathbf{v}$, $\mathbf{B}$, and $\mathbf{E}$ are acceleration, velocity, magnetic field, and electric field respectively, and are all vectorial quantities. The particle mass is m and its charge is q . For the MCX setup, we assume that a particle is azimuthally trapped in a plane midway along the axis, hereafter called the midplane, along the mirror (see Fig. 1) with no axial velocity. This assumption is justified from the theoretical prediction that axial particle losses are expected to be significantly lower than conventional mirrors when including azimuthal rotation [2]. We also consider only a radial electric field E_r , and an axial magnetic field B_z ; all other components of $\mathbf{E}$ and $\mathbf{B}$ are set to zero. This simplifies the above equation, which we may be recast as

$$\begin{aligned} \frac{1}{\Omega_i} \ddot{x} &= \frac{E_r}{B_z} \cos \theta + \dot{y} \\ \frac{1}{\Omega_i} \ddot{y} &= \frac{E_r}{B_z} \sin \theta - \dot{x}, \end{aligned} \quad (1)$$

where $\theta = \text{ArcTan}(y/x)$.

Plasmas are collisional in MCX, and plasma discharges typically span many confinement times [5]. The initial conditions are $x(0) = x_o$, $y(0) = 0$, $\dot{x}(0) = 0$, and $\dot{y}(0) = -v_o$. The velocity v_o is some fraction of $|E_r/B_z|$ (kept constant for a given simulation run), and x_o less or equal to the chamber radius R_c .

To simulate collisions, the equations are solved numerically for one collision time τ_c , the direction of travel is then randomized and used as initial conditions for $\dot{x}$ and $\dot{y}$; this process is repeated over many τ_c . For densities of $n \approx 10^{20} \text{ m}^{-3}$ and ion temperatures $T_i = 10\text{--}50 \text{ eV}$, a helium plasma with carbon or oxygen impurities would have collision times from a few to tens of microseconds [6].

Recycling is simulated with a threshold condition: if $(x^2 + y^2)^{1/2} > R_c$, then the particle is returned to a fraction

of R_c at the same theta location, and integration of Eq. 1 would be restarted for another τ_c with the randomized direction of velocities, taking $\dot{x}$ and $\dot{y}$ from the last step of the previous cycle. Note that neutral penetration depth is generally expected to be different for different ions. However, in the present simulations this penetration location was arbitrarily chosen to be $0.6 R_c$ for all ions, given the lack of neutral recycling diagnostics for MCX. Simulations with penetration models [7] for the different ions are the subject of future research. It is noted here that the present model contains no particle energy losses during collisions or recycling. On the other hand, particles experience a wide range of accelerations, and thus randomization not only occurs in the new angle for the velocity vector at each step, but also in velocity magnitude.

Figure 2 shows the result of a simulation for a singly-ionized helium ion. It is clear that after many collision times the particle has traveled through most of the domain, except for the circular region close to the axis. Simulations were also started close to the axis. In this case, the particle was pulled away by E_r from the axis within one or two collision times to the annular track region shown in Fig. 2. This indicates that the size of this exclusion zone depends mainly on E_r and Ω_i .

Comparison Between Experimental Measurements and Simulations

In order to compare particle simulations with MCX spectroscopic data, a synthetic diagnostic was created to reduce (line-average) simulated data. From a given simulation set, such as that seen in Fig. 2 at $1,387 \mu\text{s}$, the resulting data is interrogated to find the mean of all horizontal velocities v_x within the region $y \pm \delta y$. This is then compared to spectroscopic data at radius $R = y$, where the spectroscopic chord has a diameter $\approx 2 \delta y$. No additional processing, such as inversion techniques to account for luminosity profiles, was applied to the experimental or simulated data to avoid error propagation from the small number of chords used here.

The ion orbits for *He II*, *C III*, and *O II* were simulated with $E_r = (3000 \text{ V})/(0.21 \text{ m}) = 14285.7 \text{ V/m}$, $B_z = 0.2 \text{ T}$, and $R_c = 26 \text{ cm}$. The value of E_r was chosen based on the measured plasma voltage of $3.2 \pm 0.5 \text{ kV}$ on MCX for the *He* plasma discharges, and an estimate of the discharge path length across magnetic flux surfaces of 0.21 m [5]. The collision times used were: for *He II* – *He II*, $\tau_c = 11.4 \mu\text{s}$; for *He II* – *C III*, $\tau_c = 12.3 \mu\text{s}$; and for *O II* – *C III*, $\tau_c = 15.5 \mu\text{s}$. The results of the synthetic spectroscopy for each simulation is shown in Fig. 3. Note that each curve has a maximum at radii higher than 20 cm . These maxima varied depending on the recycling return fraction

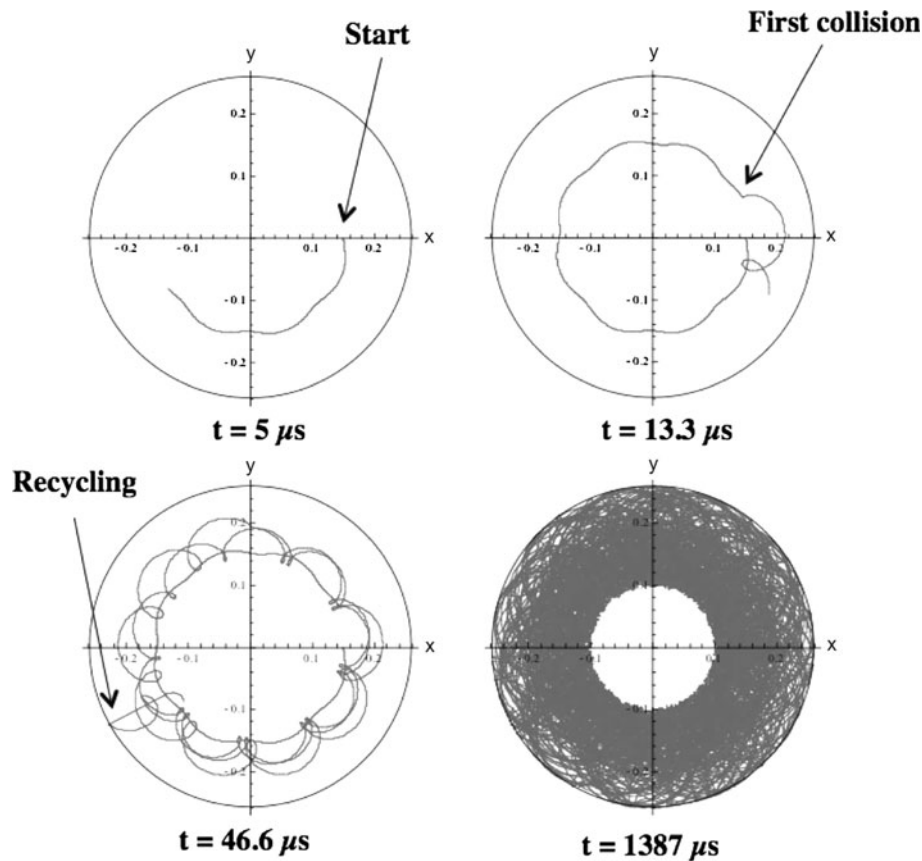


Fig. 2 Simulation of *He II* orbits. $E_r = 14285.7$ V/m, $B_z = 0.2$ T, $v_o = E_r/B_z$, $\tau_c = 11.4$ μ s, $R_c = 0.26$ m. The circumference represents the MCX vessel

mentioned in “Particle Orbit Model”; higher return fractions would yield lower maxima radii.

At MCX we have the ability to obtain five simultaneous spectroscopic measurements using a McPherson (Turner-Czerny type) spectrometer with a Stanford Computer Optics high-speed intensified camera. For the present comparison, the chords were arranged as shown in Fig. 4.

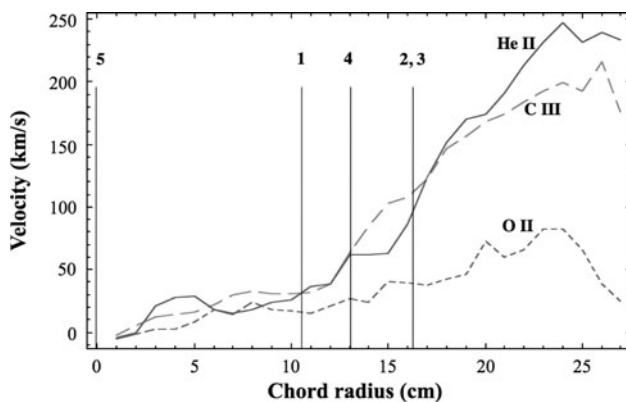


Fig. 3 Average v_x along constant y chords of the simulated data sets for *He II*, *C III*, and *O II*. Vertical lines correspond to radial chords indicated in Fig. 4

All plasma discharges were prefilled with helium at 5 mTorr. The spectrometer was calibrated using a neon lamp in the 464–476 nm range prior to obtaining the data, and none of the mechanical systems were moved thereafter. At every plasma discharge, a single image of the five chords would be obtained with a shutter integration time of 100 μ s. The following lines were identified in the calibrated wavelength range: *He II*, 468.57 nm; *C III*, 464.742 nm; *O II*, 464.18104 nm.

Velocities measured from Doppler shifts are shown in Fig. 5 together with simulation data obtained from Fig. 3. The error bars represent one standard deviation from the mean of four shots (numbered 21–24 taken on February 1, 2010), and include shot to shot variations as well as difficulty identifying the true peak in the spectroscopic data. From the three peaks discussed here, *He II* has the lowest signal to noise ratio, which is another cause for the large error bars. (In previous experimental runs, an increased number of shots would only reduce the error bars by a small amount.)

It was recognized that even after calibration, either electrical noise or mechanical vibrations affect the spectrometer at the time of each plasma discharge. For this

Fig. 4 Schematic diagram of spectroscopy fibers on MCX. Fiber radius is the distance from the MCX vessel axis to the perpendicular of each chord (each represented by an arrow)

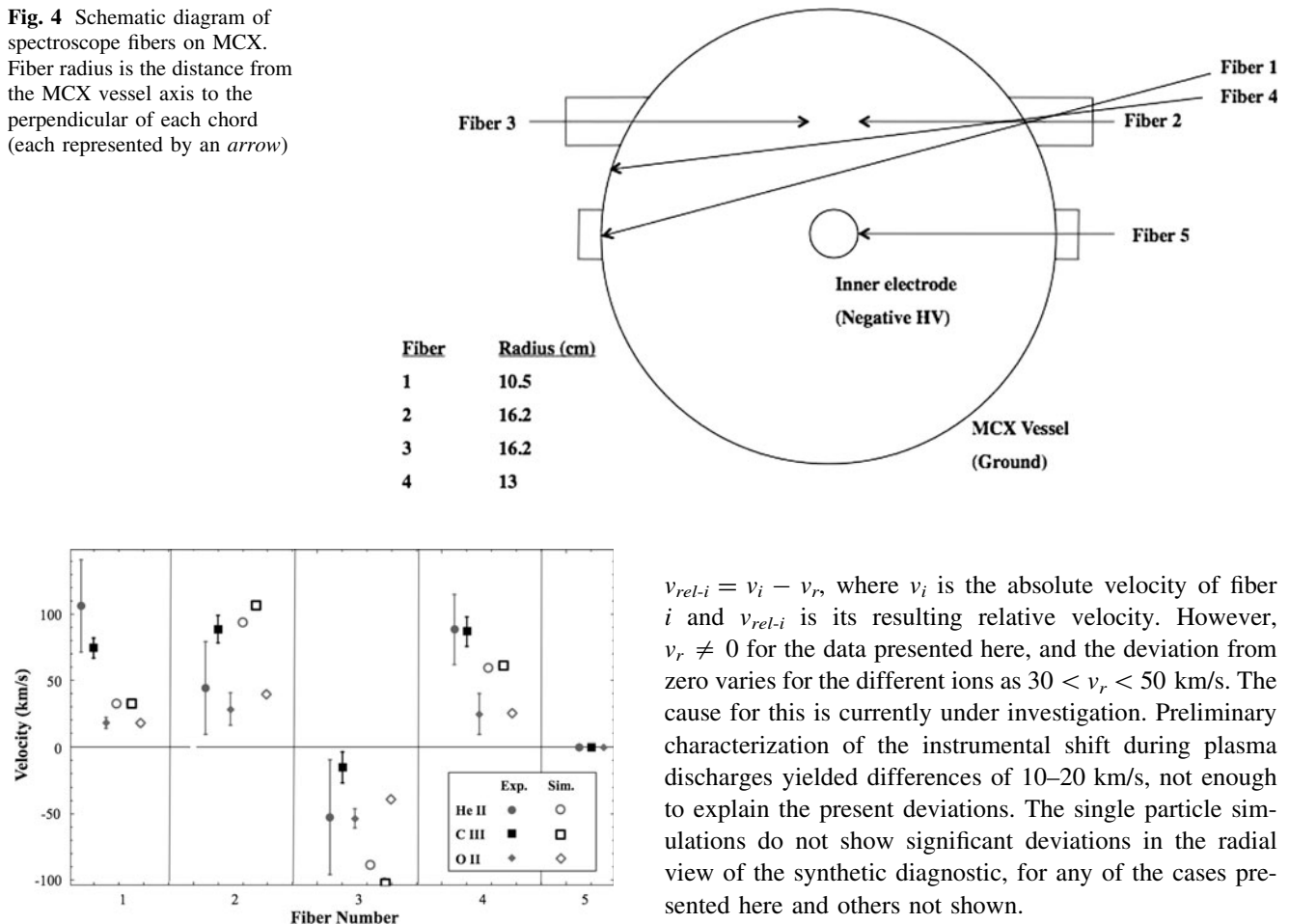


Fig. 5 Measured and simulated velocities at the radii indicated in Fig. 4. Error bars for the measured velocities are for v_{rel-i} (see text for details)

reason the velocities shown in Fig. 5 are subtracted from those measured at the radial view (Fiber 5) which in principle should have zero velocity. This is not always the case (see below). The error bars for the radial chord deviations (not shown) are similar to those of other chords.

Discussion

The experimental velocity data indicates that different ions rotate at different speeds at different radial locations. In particular, *O II* is consistently slower than *He II* and *C III*. This is in qualitative agreement with the present simulations. Figure 5 also shows that the anti-parallel views (fibers 2 and 3) do not yield symmetric velocities. As mentioned in the previous section, velocities are with respect to the measured velocity in the radial view, v_r . At every plasma discharge and for chords 1 through 4,

$v_{rel-i} = v_i - v_r$, where v_i is the absolute velocity of fiber i and v_{rel-i} is its resulting relative velocity. However, $v_r \neq 0$ for the data presented here, and the deviation from zero varies for the different ions as $30 < v_r < 50$ km/s. The cause for this is currently under investigation. Preliminary characterization of the instrumental shift during plasma discharges yielded differences of 10–20 km/s, not enough to explain the present deviations. The single particle simulations do not show significant deviations in the radial view of the synthetic diagnostic, for any of the cases presented here and others not shown.

One hypothesis being considered is the existence of asymmetries in the plasma. Since the deviations from zero for v_r are reproducible within the aforementioned error bars, these asymmetries would have to appear consistently at every shot, within 100 μ s. Within one experimental day (typically more than 20 plasma discharges) v_r is either only positive or only negative if experimental parameters are left unchanged. We have yet to investigate variations in experimental parameters such as discharge voltage or prefill pressure, versus v_r . Doppler shift measurements at several simultaneous radial views are planned for future experimental campaigns.

In the present measurements, plasma voltage would always exhibit a sawtooth-type oscillation of a few kilohertz (with the phase of the oscillations varying from shot to shot). This is typical for MCX helium discharges where the capacitor bank that provides power to create and sustain the plasmas is charged to 10 kV or more. The present data was taken with a charge voltage of 11 kV (plasma discharge duration of 5 ms). It was at this setting that the *He II* 468.57 nm line would appear significantly above the noise. The electron temperature for the present plasmas was estimated from ionization balance modeling [8] to be $T_e > 5$ eV.

The particle simulations have yielded interesting qualitative insight into orbit evolution from collisions. However, the model used in the computations contains many simplifications that may ultimately affect the veracity of the results. E_r , for example, is treated as constant whereas the real field may have a decreasing profile near the vessel walls. Collisionality and recycling may also vary with radial location and temperature. Improvements to the model and further comparisons to experimental data are ongoing and will be reported elsewhere.

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References

1. R.F. Ellis, A. Case, R. Elton, J. Ghosh, H. Griem, A. Hassam, R. Lunsford, S. Messer, C. Teodorescu, *Phys. Plasmas* **12**, 055704 (2005)
2. A.B. Hassam, *Comments Plasma Phys. Control. Fusion* **18**, 263 (1997)
3. A.B. Hassam, *Phys. Fluids B* **4**, 485 (1992)
4. J. Ghosh, R.C. Elton, H.R. Griem, A. Case, R. Ellis, A.B. Hassam, C. Teodorescu, S. Messer, *Phys. Plasmas* **11**, 3813 (2004)
5. C. Teodorescu, R. Clary, R.F. Ellis, A.B. Hassam, R. Lunsford, I. Uzun-Kaymak, W.C. Young, *Phys. Plasmas* **15**, 042504 (2008)
6. J.D. Huba, *NRL plasma formulary*. Naval Research Laboratory, Washington, pp. 31–39 (2007)
7. S.W. Ng, A.B. Hassam, *Phys. Plasmas* **14**, 102508 (2007)
8. Y. Ralchenko, Personal communication, March 17 (2010)